GateTunnel

System Description

Abstract
This document describes the GateTunnel support system of the Eclipse Arrowhead framework (documentation edition 5.3.0), as implemented in `ah5-GateTunnel-java-spring` with an independent local-cloud deployment definition in its own folder.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the GateTunnel system. GateTunnel carries the inter-cloud service calls negotiated by the Gatekeepers: a consumer session listens locally and forwards each request over the relay to the provider cloud's GateTunnel, which calls the provider.

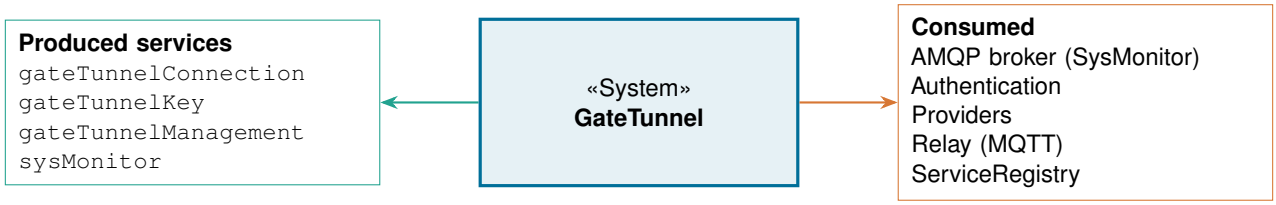


Figure 1: SysML-style block view of GateTunnel with its produced and consumed services.

The rest of this document is organized as follows. In Section 2.1, we reference major prior art capabilities of the system. In Section 2.2, we describe the intended usage of the system. In Section 2.3, we describe fundamental properties provided by the system. In Section 2.4, we describe delimitations of capabilities of the system. In Section 3, we describe the abstract service functions consumed or produced by the system. In Section 4, we describe the security capabilities of the system.

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Gateway; native Java port of the former Python implementation (Python sources kept for rollback).

2.2 How This System Is Meant to Be Used

The local Gatekeeper opens provider and consumer sessions; consumers call the consumer-session endpoint returned by orchestration.

2.3 System functionalities and properties

2.3.1 Functional properties of the system

- Relay sessions restricted to peer, consumer and negotiated operations.
- Legacy control-session API.
- Session management.

2.3.2 Configuration of system properties

The system is configured with environment variables. The variables that shape its behaviour are:

Property	Default	Meaning
`PORT`	8453	HTTPS port.
`MIN_TUNNEL_PORT` / `MAX_TUNNEL_PORT`	8000 / 8099	Session ports.
`ADVERTISED_ADDRESS`	gatetunnel	Address returned to consumers.
`RELAY_HOST` / `RELAY_PORT`	arrowhead-relay / 8883	Relay.
`NEIGHBOR_CLOUDS` / `FEDERATION_PEER_ORGANIZATIONS`	GoCloud / (empty)	Trusted peers.
`SESSION_MAX_AGE_SECONDS` / `SESSION_IDLE_SECONDS`	43200 / 3600	Session lifetimes.
`FEDERATION_PROFILE`	legacy-mtls	Federation profile.

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

2.3.3 Data stored by the system

Control sessions (SQLite); relay sessions in memory.

2.3.4 Non functional properties

Bounded concurrent requests per session; 12 h maximum and 1 h idle lifetime.

2.3.5 Stateful or stateless

Consumer relay sessions are volatile and renegotiated after restart.

2.4 Important Delimitations

Not end-to-end encrypted under `legacy-mtls`. The legacy control sessions have no data plane.

3 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

3.1 Produced services

Service	Version	Interfaces	SD	IDD
`gateTunnelConnection`	5.2.2	HTTP(S)	[1]	[2]
`gateTunnelKey`	5.2.2	HTTP(S)	[3]	[4]
`gateTunnelManagement`	5.2.2	HTTP(S)	[5]	[6]
`sysMonitor`	1.0.0	AMQP(S)	[7]	[8]

All services use `generic_https` with `CERT_AUTH`; `sysMonitor` uses `generic_amqps`. The container also answers `GET /health` on a loopback-only plain-HTTP port for the container health check.

3.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	serviceDiscovery	register	Registration.	always
Authentication	identity	identity-login, identity-verify	Own token; caller verification.	always
Relay (MQTT)	inter-cloud messages	forwarded requests	Peer GateTunnels.	sessions
Providers	negotiated service	negotiated operations	Forwarded calls.	provider sessions
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	always

Only Authentication's `identity` and the orchestrator are looked up in the ServiceRegistry; other services are reached through DynamicServiceOrchestration (results cached), and every such call needs a Consumer-Authorization grant.

4 Security

The standalone Java system requires HTTPS with mutual TLS and X.509 certificates following the Arrowhead system certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`, DNQ `sy`). There is no insecure runtime mode. Identity tokens are verified through Authentication and bound to the caller certificate. A separate loopback health endpoint is infrastructure only. Provision credentials and a valid CRL using the project-local environment template and Compose definition.

4.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; SQLite storage is local to this system's private volume. Passwords are read from Compose file secrets; certificate and private-key files are mounted individually.
- **System authentication:** mutual TLS with a system or operator certificate of the cloud and an identity token of the same system, verified with Authentication; declared identities are not accepted. Client certificates are checked against the CRL.
- **Authorization of produced services:** per operation by caller lists in the implementation and, where stated, by ConsumerAuthorization.
- Only the Gatekeeper opens relay sessions; requests outside the negotiated operations are refused (403).

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

5 References

[1] J. Delsing, "gateTunnelConnection – Service Description (GateTunnel)," Eclipse Arrowhead, SD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelConnection-SD.pdf.

[2] ——, "gateTunnelConnection HTTP/TLS/JSON – Interface Design Description (GateTunnel)," Eclipse Arrowhead, IDD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelConnection-HTTP-IDD.pdf.

[3] ——, "gateTunnelKey – Service Description (GateTunnel)," Eclipse Arrowhead, SD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelKey-SD.pdf.

[4] ——, "gateTunnelKey HTTP/TLS/JSON – Interface Design Description (GateTunnel)," Eclipse Arrowhead, IDD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelKey-HTTP-IDD.pdf.

[5] ——, "gateTunnelManagement – Service Description (GateTunnel)," Eclipse Arrowhead, SD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelManagement-SD.pdf.

[6] ——, "gateTunnelManagement HTTP/TLS/JSON – Interface Design Description (GateTunnel)," Eclipse Arrowhead, IDD 5.3.0, 2026, gateTunnel-v5.3.0-gateTunnelManagement-HTTP-IDD.pdf.

[7] ——, "sysMonitor – Service Description (GateTunnel)," Eclipse Arrowhead, SD 5.3.0, 2026, gateTunnel-v5.3.0-sysMonitor-SD.pdf.

[8] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (GateTunnel)," Eclipse Arrowhead, IDD 5.3.0, 2026, gateTunnel-v5.3.0-sysMonitor-AMQP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Independent Java build, private helpers and state, per-system deployment and configuration.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)
4	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	